The modulus of a number

Distance from zero — one idea that turns every modulus equation and inequality into something you can already solve.

LESSON 53–54 | 2 × 40 MIN | ALGEBRA 8, §17 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define $|a|$ and evaluate it.
- Read $|x|$ as the distance from x to 0 on the number line.
- Solve equations of the form $|x - a| = b$.
- Solve inequalities $|x| < b$ and $|x| > b$.

TEXTBOOK

National	§17, pp. 105–110
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The definition

DEFINITION

$$|a| = a \text{ if } a \geq 0, \quad |a| = -a \text{ if } a < 0$$

So $|5| = 5$, $|-5| = 5$, $|0| = 0$. The modulus is **never negative**.

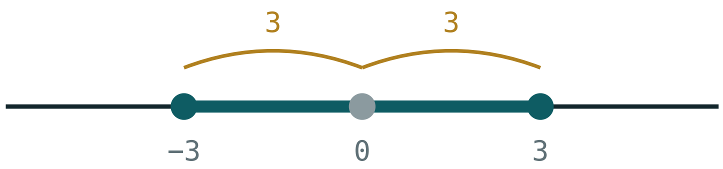
−A IS NOT A NEGATIVE NUMBER

If $a = -5$ then $-a = 5$. The second line of the definition *removes* the minus sign; it does not add one.

Two facts worth stating at once: $|a| \geq 0$ always, and $|a| = |-a|$ — opposite numbers have the same modulus.

The picture: distance from zero

$|a|$ is the **distance** from the point a to the origin. That single sentence solves the whole topic.



| x | ≤ 3 means −3 ≤ x ≤ 3

Every point within 3 of the origin satisfies $|x| \le 3$ — the segment from −3 to 3.

More generally, $|x - a|$ is the distance from x to a. So $|x - 4| = 3$ asks: which points are exactly 3 away from 4? Answer: 1 and 7.

Equations

RULE

For $b > 0$: $|A| = b$ means $A = b$ **or** $A = -b$.
$|A| = 0$ means $A = 0$. $|A| = b$ with $b < 0$ has **no solutions**.

$$|x - 4| = 3 \Longrightarrow x - 4 = 3 \text{ or } x - 4 = -3 \Longrightarrow x = 7 \text{ or } x = 1$$

Always check the sign of the right-hand side first. $|x + 2| = -5$ needs no working at all — a modulus cannot be negative.

Inequalities

TWO SHAPES, TWO ANSWERS

$$|x| < b \text{ means } -b < x < b \text{ — one interval, in the middle}$$

$$|x| > b \text{ means } x < -b \textbf{ or } x > b \text{ — two rays, outside}$$

The picture makes it obvious: “less than b from zero” is the middle piece; “more than b from zero” is everything outside.

Same with a shift: $|x - 5| \le 2$ means the distance from 5 is at most 2, so $3 \le x \le 7$.

02 Worked examples

EXAMPLE 1 *Solve* $|2x - 1| = 7$

1. $2x - 1 = 7$ or $2x - 1 = -7$

   Two cases.

2. $2x = 8$, so $x = 4$

   First case.

3. $2x = -6$, so $x = -3$

   Second case.

4. *check:* $|7| = 7$ ✓ *and* $|-7| = 7$ ✓

ANSWER

$x = 4$ or $x = -3$

EXAMPLE 2 *Solve* $|x - 3| < 5$

1. The distance from 3 is less than 5.

   Read it as distance.

2. $-5 < x - 3 < 5$

   The middle-piece form.

3. $-2 < x < 8$

   Add 3 throughout.

ANSWER

$(-2; 8)$

EXAMPLE 3 ***Solve* $|x + 1| \geq 4$**

1. The distance from -1 is at least 4.
   This is the "outside" shape.

2. $x + 1 \leq -4$ or $x + 1 \geq 4$
   Two rays.

3. $x \leq -5$ or $x \geq 3$

ANSWER

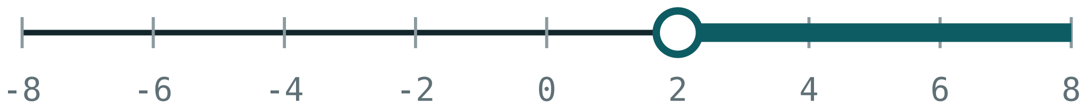
$(-\infty; -5] \cup [3; +\infty)$

03 Interactive model

Set the boundary at 3 and switch between < and > — the two answer shapes appear side by side.

Inside or outside?

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Modulus (absolute value)	Modul (absolyut qiymat)	Модуль (абсолютная величина)
Distance from zero	Noldan uzoqlik	Расстояние от нуля
Non-negative	Manfiy bo‘lmagan	Неотрицательный
Two cases	Ikki hol	Два случая
Opposite numbers	Qarama-qarshi sonlar	Противоположные числа
Equation with a modulus	Modulli tenglama	Уравнение с модулем
Inequality with a modulus	Modulli tengsizlik	Неравенство с модулем
Solution set	Yechimlar to‘plami	Множество решений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$|-7|$ equals:

- -7
- 7
- 0
- ± 7

$|x| = 5$ gives:

- $x = 5$
- $x = -5$
- $x = 5$ or $x = -5$
- no solutions

$|x| < 4$ means:

- $x < 4$
- $-4 < x < 4$
- $x > 4$
- $x < -4$ or $x > 4$

$|x + 2| = -3$ has:

- one solution
- two solutions
- no solutions
- every number

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find* $|\,9\,|$

   ANSWER 9

2. *Find* $|-9\,|$

   ANSWER 9

3. *Find* $|\,0\,|$

   ANSWER 0

4. *Find* $|-3\,| + |\,4\,|$

   ANSWER 7

5. *Solve* $|\,x\,| = 6$

   ANSWER $x = 6$ or $x = -6$

6. *Solve* $|\,x\,| = 0$

   ANSWER $x = 0$

7. *Solve* $|\,x\,| < 2$

   ANSWER $-2 < x < 2$

Medium · the standard question

7 PROBLEMS

1. *Solve* $|\,x - 4\,| = 3$

   ANSWER $x = 7$ or $x = 1$

2. *Solve* $|\,2x - 1\,| = 7$

   ANSWER $x = 4$ or $x = -3$

3. *Solve* $|\,x + 2\,| = 5$

   ANSWER $x = 3$ or $x = -7$

4. *Solve* $|\,x - 3\,| < 5$

   ANSWER $(-2; 8)$

5. *Solve* $|\,x\,| \geq 3$

   ANSWER $x \leq -3$ or $x \geq 3$

6. *Solve* $|\,x + 1\,| \geq 4$

   ANSWER $x \leq -5$ or $x \geq 3$

7. *Solve* $|\,x + 2\,| = -5$

   ANSWER No solutions.

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $|3x + 6| = 12$

   ANSWER $x = 2$ or $x = -6$

2. *Solve* $|x - 5| \leq 2$

   ANSWER $[3; 7]$

3. *Solve* $|2x - 3| < 5$

   ANSWER $-1 < x < 4$

4. *Solve* $|1 - x| = 4$

   ANSWER $x = -3$ or $x = 5$

5. *How many whole numbers satisfy* $|x| \leq 4$?

   ANSWER Nine: -4 to 4.

6. *Solve* $|x - 2| > 0$

   ANSWER Every x except $x = 2$.

7. *Find the distance between the points* $x = -3$ *and* $x = 8$ *using a modulus*

   ANSWER $|8 - (-3)| = 11$

07 Homework

Homework — 6 problems

Algebra 8, §17, pp. 105–110. Draw the number line for the inequalities.

1. *Find* $|-12|$, $|12|$ *and* $|-12| - |5|$
2. *Solve* $|x - 6| = 4$
3. *Solve* $|3x + 1| = 10$
4. *Solve* $|x| \leq 7$
5. *Solve* $|x - 1| > 3$

6. *Explain why* $|x + 3| = -2$ *has no solutions*